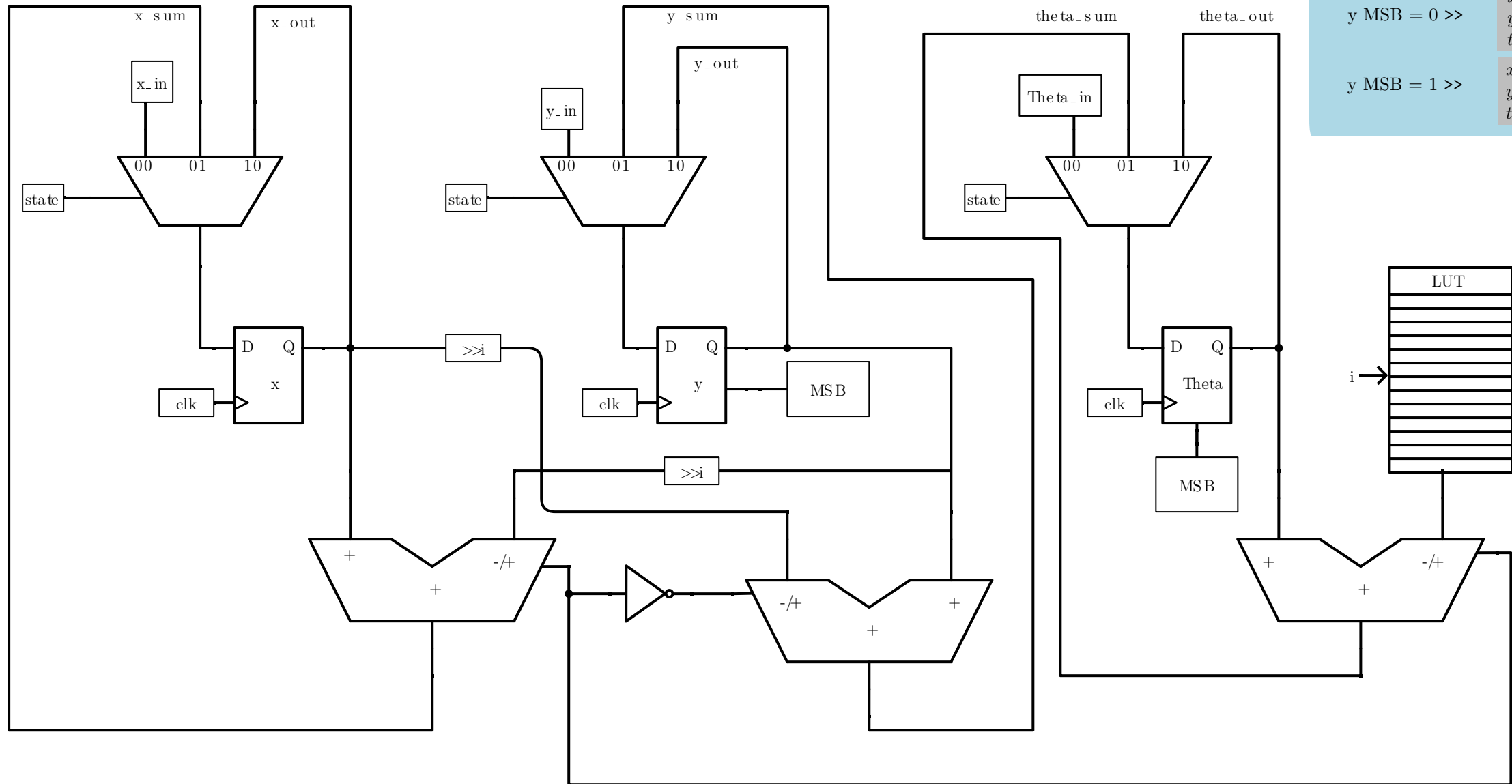
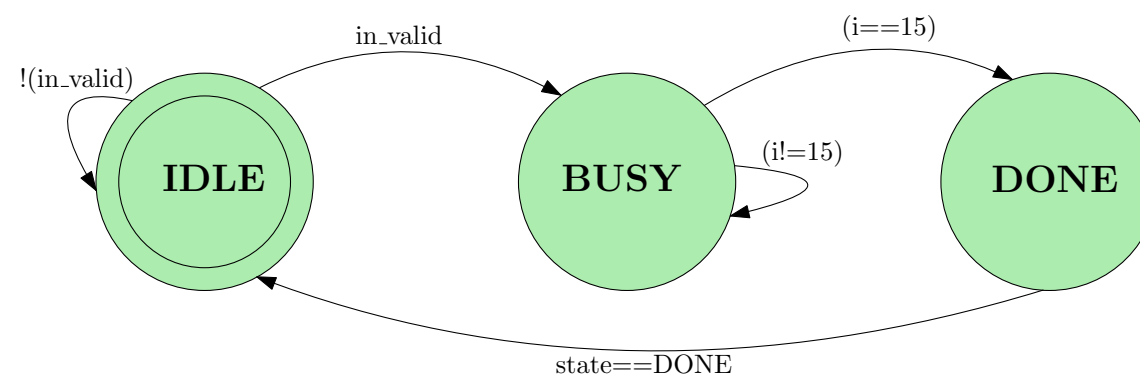


CORDIC ROLLING ARCHITECTURE

```
mode = 0 : cosine, sine (Rotation Mode)
mode = 1 : Magnitude, Angle (Vectoring Mode)
>>i : Barrel Shifter
```



Control FSM



mode = 0

$$\begin{aligned} \text{Theta MSB} = 0 &>> \\ x_{k+1} &= x_k - 2^{-k} * y_k \\ y_{k+1} &= y_k + 2^{-k} * x_k \\ \text{theta}_{k+1} &= \text{theta}_k - \text{angle} \end{aligned}$$

Theta MSB = 1 >>	$x_{k+1} = x_k + 2^{-k} * y_k$ $y_{k+1} = y_k - 2^{-k} * x_k$ $theta_{k+1} = theta_k + angle$
------------------	---

mode = 1

y MSB = 0 >>

$$\begin{aligned} x_{k+1} &= x_k + 2^{-k} * y_k \\ y_{k+1} &= y_k - 2^{-k} * x_k \\ \theta_{k+1} &= \theta_k + \text{angle} \end{aligned}$$

y MSB = 1 >>

$$\begin{aligned} x_{k+1} &= x_k - 2^{-k} * y_k \\ y_k &= y_k + 2^{-k} * x_k \\ \text{theta}_{k+1} &= \text{theta}_k - \text{angle} \end{aligned}$$

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